

Question	Answer	Marks
3(a)	10.0 cm	A1
3(b)(i)	$v = f\lambda$ $= 16 \times 0.40$	C1
	$= 6.4 \text{ m s}^{-1}$	A1
3(b)(ii)	(X and Y have a) <u>constant</u> phase difference (of 180°) so (they are) coherent	B1
3(c)	A single wave of amplitude 10.0 cm	B1
	A single negative sine wave of wavelength 0.40 m	B1
3(d)	$I \propto A^2$ $\frac{I_X}{I_Y} = 10^2 / 20^2$	C1
	ratio = 0.25	A1

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